PHASE 1

Executive Summary and Technology Stack

Product summary and the technologies behind ComicCraft

COMICCRAFT • AI COMIC STORY CREATOR • TECHNICAL DOCUMENTATION

1. Executive Summary

ComicCraft is a full-stack web application that turns a single-line story idea into a complete, illustrated comic. A user supplies a prompt, a character name, a setting, a tone and an art style; the application then generates a structured panel outline, writes narration, captions and dialogue, illustrates every panel with an AI image model, renders an in-browser comic reader, and exports a print-ready PDF.

The system is built on FastAPI with Pydantic v2 for validation, Jinja2 for server-rendered templates, a two-stage Gemini text pipeline (Flash for outlining, Pro for narrative writing), a pluggable image generation layer (local Stable Diffusion, the Hugging Face Inference API, or Gemini image generation), and FPDF2 for PDF export. The frontend is deliberately framework-free — mobile-first CSS and vanilla JavaScript — to keep the client lightweight and dependency-free.

1.1 Purpose of this Document

This document is a professional technical reference for developers, reviewers and stakeholders. It covers the system architecture, environment configuration, installation, the HTTP API, the core application source code, automated testing, troubleshooting, and guidance for extending the platform.

2. Technology Stack

Layer	Technology
Backend framework	FastAPI + Uvicorn (ASGI)
Data validation	Pydantic v2
Templating	Jinja2 (server-rendered HTML)
Outline model	Gemini Flash — GEMINI_OUTLINE_MODEL
Story model	Gemini Pro — GEMINI_STORY_MODEL
Illustration	Stable Diffusion (Diffusers), Hugging Face Inference API, or Gemini image model
PDF export	FPDF / fpdf2
Frontend	Mobile-first CSS + vanilla JavaScript (no framework)
Testing	pytest + httpx (TestClient)

Design principle
Every model name, API key and provider choice is environment-driven (.env). Nothing is hard-coded, so text models, image providers and generation limits can be swapped without touching application code.